Where Did This Code Come From? Detecting Assistant-Pasted Code in Student Editor Logs

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Abstract

When a course permits an AI assistant, instructors still want to know how much of a submission a student wrote. We study paste events in an instrumented editor used by 296 CS1 students alongside an integrated assistant. A classifier using paste size, time since the last assistant response and token overlap with that response identified assistant-originated pastes with F1 = 0.88 against assistant logs. Across the term, 23% of final submissions contained more than half pasted assistant code, concentrated in the last 24 hours before deadlines. We discuss uses of provenance signals that support learning rather than policing.

1 Introduction

Integrated assistants make provenance visible to the platform but not to the instructor. Paste events are a natural boundary: a student who copies from an assistant panel or another window produces a single large insertion rather than a stream of keystrokes. We ask how well paste events reveal assistant origin and what the pattern of pasting looks like over a term.

2 Data

The course used a browser IDE with an assistant side panel. With consent (296 of 341 students), the IDE logged insertions, deletions, pastes, runs and assistant exchanges. Ground truth for assistant origin came from the assistant log: a paste was labeled assistant-originated if at least 80% of its tokens appeared in an assistant response from the previous 10 minutes.

3 Detector

We trained a logistic regression on features that an instructor-facing tool could compute without the assistant log: paste length in tokens, seconds since the last assistant response, whether the paste contained comments in the assistant's style ("# Step 1:"), and indentation consistency with surrounding code.

Feature set	Precision	Recall	F1
Paste length only	0.61	0.79	0.69
+ time since assistant response	0.83	0.84	0.83
+ style and indentation	0.87	0.89	0.88

Time since the assistant response carried most of the signal: 71% of assistant-originated pastes happened within 20 seconds of a response.

4 Patterns over the term

Pasting rose across the term, from 9% of submissions with majority-pasted code in week 2 to 31% in week 12. Majority-pasted submissions clustered near deadlines: 64% were finalized in the last 24

hours. Students who pasted heavily early in the term did not score lower on the in-person midterm once prior programming experience was controlled, but students whose pasting increased sharply after week 6 did (by 7.4 points on average).

5 Discussion

Provenance can be used to punish or to teach. We argue for the second: showing a student the share of their submission that came from the assistant, privately, before the deadline, and pairing it with an invitation to explain the pasted code. Our detector is too imperfect for integrity cases and we do not recommend using it for them.

6 Limitations

One course, one assistant. Students can retype assistant code, which our detector misses by design.

References

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